Boreas - Architecture, Design and Tradeoffs

Product

Boreas is a day-ahead GB national wind desk for January 2026.
It shows the WINDFOR run a trader could use before the EPEX gate,
pairs it to FUELHH WIND actuals, and reports MAE/RMSE/MAPE/bias plus a P90 residual band.

Architecture

React SPA (Vite) -> FastAPI Boreas API -> Elexon Insights stream (FUELHH, WINDFOR).
Local: docker compose (web:8080 proxies /api to api:8000).
Deploy target: Vercel static frontend + Python serverless (Mangum).
Cache: data/cache/jan2026_bundle.json baked for cold starts.

ASCII diagram:

```
[ React SPA ] --/api/*--> [ FastAPI ] --HTTPS--> [ Elexon Insights ]
     |                       |
  nginx (docker)     JSON cache volume
```

Decision-window rule

For delivery day D (00:00-24:00 Europe/London):
1. Gate close = 09:20 CET on D-1 (EPEX SPOT GB day-ahead).
2. Eligible forecasts: WINDFOR publishTime <= gate.
3. Take the latest eligible publishTime.
4. Keep points with startTime on day D.
5. Pair to FUELHH fuelType=WIND on matching startTime.
Note: WINDFOR is hourly; FUELHH is half-hourly -> 744 paired intervals in Jan 2026.

Tradeoffs

Stream dataset APIs vs opinionated forecast endpoints
Chosen streams for publish-time history required by the brief.
Baked cache vs live-only
Deterministic demos and Vercel cold starts; refresh via POST /api/refresh.
P90 abs error as residual band
Non-parametric and procurement-actionable.
Vercel-only (no Heroku)
Per delivery constraint; Docker is the guaranteed local path.

Security and ops

Elexon Insights is public (no key). Optional ELEXON_API_KEY for future gateways.
No PII; national aggregates only.